PROFESSIONAL BRIEFINGS . . .

Measuring IS: How Does Your Organization Rate?

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This is the first of a series of briefings and tutorials that will appear from time to time in future issues of The DATA BASE for Advances in Information Systems.

Introduction

In today's difficult economy, companies are re-organizing and streamlining wherever possible in order to save money and attract customers. This new corporate emphasis on better organizational performance, i.e., both productivity and quality goods and services, challenges all parts of an organization, including IS, to make improvements in what they do and how they operate. This is easier said than done and many organizations are having difficulties knowing where to start, what to do, and what their goals should be. Organizational learning and change are the hallmarks of business in the nineties and the starting point for this effort is the collection of accurate and adequate information and its appropriate analysis.

It is not surprising therefore that comparison, measurement, and evaluation have become something of a preoccupation in many businesses in recent years. The U.S. National Quality Institute, (which establishes the detailed criteria for the Malcolm Baldridge Awards), suggests there are three components of effective information and analysis which act as the "brain center" driving the corporate improvement effort, regardless of a company's organization or structure (Baldridge Criteria, 1993):

1. Scope and management of quality and performance information.
2. Collection, analysis, and use of company level data, including customer data and operations data, and linking performance data to overall financial performance.
3. Competitive comparisons and benchmarking.

These criteria can also be used by individual corporate subunits to determine how well they contribute to overall quality and performance.

One organizational subunit receiving a considerable amount of executive attention in this regard is the information systems (IS) function. Many companies are looking to information technology to help them support corporate restructuring, but to do it with increasingly fewer resources. Unfortunately, IS has also been an area of the company that has been extremely difficult to measure and evaluate. IS also continues to have a credibility problem in some organizations where executives believe that IS has contributed little or nothing to the corporate bottom line (Roach, 1989). As well, there are still many users who tend to feel that IS service is lacking or could be bought cheaper elsewhere. With IS budgets coming under closer and closer scrutiny, IS itself is placing new emphasis on measurement to demonstrate its contribution to overall corporate performance, both through quality services and systems and through its ability to do so cost-effectively.

This paper looks at the three components of information and its analysis, as outlined above, from the perspective of the IS subunit. It summarizes what IS departments are doing to assess and analyze their organizations and makes suggestions about how the measurement and evaluation of IS both internally and externally could be improved.

Methodology

In order to explore and assess what IS organizations are doing to evaluate and compare themselves, the authors convened a day-long focus group of senior IS managers from ten leading Canadian firms. Five industry sectors were represented (see Table 1). Before the sessions, each group member was asked to prepare a twenty-minute presentation examining the measures their organization used to understand and assess their own procedures and performance and how they compared their overall performance to others.

Industry	Number
Retail	2
Manufacturing	2
Insurance	3
Banking	2
Telecommunications	1
Total	10

Table 1. Industries Participating in the Measurement Focus Group

The following questions were given to the focus group participants as a guide:

1. What indicators/measures do you track/monitor in IS
   a) on an individual basis?
   b) on a project basis?
   c) on a department basis?
   Which ones are the best/most meaningful from your point of view?
2. Do you compare yourself now to other IS organizations and the companies they belong to? If so, on what basis and to whom?
3. What indicators/measures are reported to senior executives concerning IS?
4. What indicators/measures are reported to users on IS?
5. What would you like to know about your own IS organization? Why?
6. What indicators would senior management or users like to know about IS? Why would they like it?
7. What would you like to know about other IS organizations in your industry?
8. What would you like to know about other IS organizations in general?

During the day, each participant answered questions about their presentations. Extensive discussions followed. In addition, written copies of each presentation and supporting documentation (e.g., survey instruments, sample reports to management, descriptions of measures used) were provided to the authors who were the focus group organizers. All participants took part in determining the group's final recommendations for other managers.

After this paper was drafted, it was reviewed by each participant for accuracy and completeness. Each participant was also polled to determine if the paper as a whole represented a fair assessment of their experiences and needs pertaining to IS measurement. Some wording changes were suggested, but all members agreed that the paper here presented fairly sumarizes the discussion and conclusions of the focus group session.

Scope and Management of IS Information

Group members indicated that they had collected a wide assortment of statistics and qualitative data for their own internal use. The list in Appendix A gives a compilation of measures they collect regularly. Each organization has its own set of measures and comparative practices which bear little resemblance to any others. Some make customer satisfaction surveys a focal point of their efforts, while others do little or nothing in this area. Some have extensive and comprehensive programs of evaluation; others are relatively haphazard about data collection. Some relate measures to bottom line benefits; others track individual statistics only, without linking them to any larger context.

Accurate and appropriate data are a prerequisite for any successful internal or external program of measurement and evaluation. Current IS measurement practices were cited as being problematic in three ways:

1. **Information Gaps.** Most organizations do *not* track progress in one or more of the following key areas essential to monitoring quality and operational performance improvement (Baldridge Criteria, 1993):
   - Product and service performance (i.e., systems and operations)
   - Internal operations and performance
   - Supplier performance
   - Cost and financial information.

2. **Lack of Standards.** IS measurement suffers from a lack of standards in almost all areas. Current measurement practices are characterized by a general lack of agreement about what constitutes appropriate data. As a result, with the exception of some performance measures in operations, very few statistics are comparable from organization to organization.

3. **Difficulty in Data Collection.** Much data collection in IS organizations imposes a serious burden on the staff already under pressure to complete their work. As a result, accuracy and scope of data collection are often hampered. For example, determining the function points associated with a system to estimate a system's benefits is an extraneous task that does nothing to further the development process.

Altogether, the gaps in what is measured, the lack of standardized measures, and the difficulty of producing measures suggest that IS organizations need to examine seriously what they measure and how they do it in order to obtain the type of data they need to improve their performance and quality and to make it demonstrable to their business management.

Linking Data to Performance

IS functions need to measure their performance for three reasons. First, such information can be an aid to credibility reassuring senior managers that their IS function is performing efficiently and effectively and is continually improving. Second, it is an aid to productivity improvement, demonstrating trends over time and identifying areas where improvement is necessary. Finally, it can be the catalyst to organizational transformation, supporting company and functional review, action, and planning. The key to achieving each of these is demonstrating the link between the data collected and functional and corporate performance.

This link can be made most easily at present in the operational performance measures many companies collect. Response time figures, turnaround time, workload figures, and recovery times can be monitored over time and are important reflections of service to the company. Each of these, in turn, relate to broader measures of organizational performance, since operational systems support so many corporate functions. For example, monitoring calls to a help desk can identify the quality of the services provided and can highlight important operational and system problems.

Unfortunately, all too often these measures are collected and used only in isolation. One large company is proud of its efficient help desk but fails to analyze the problems reported to prevent them from recurring! Other companies monitor system availability statistics to two decimal places for each piece of their network — but take no action on the numbers!

However, because for the business "availability" is a function of all parts of the network working together, these do not reflect the actual availability of computer services to users. Thus, while IS managers collect a large amount of data that accurately measure the efficiency of IS operations, they often neglect to interpret these data to make the necessary linkages with IS performance and quality. Additional analysis of these measures could yield more valuable information about the IS function and its impact on corporate performance.

In systems development, the link between measures and performance becomes even more tenuous. All too often, in IS as in other parts of the organization, measures assess what *can be* measured rather than what *should be* measured. Organizations then often end up collecting data about the wrong things and the resulting information can actually detract from overall effectiveness simply by putting management's focus on the wrong things (Goldratt and Cox, 1984). For example, measures such as return on equity (ROE) or return on assets (ROA) for a project, cannot reflect the many and varied impacts systems can have on an organization and may result in the wrong systems as a whole developed (Markus and Soh, 1993).

Problems with the meaningfulness of some of the measures used also prevent clear linkages with performance from being made. Do executives care that dollar sales per gigabyte of storage are increasing? What does this mean to the company's bottom line? Business people often complain that IS statistics tell them little about what they really need to know. For example, chargeout rates broken down by type of hardware and software may be meaningful to IS professionals but to business people they are not figures that are intuitively easy to understand or that translate onto any yardstick with which they are familiar.

Even simple descriptive measures can be difficult to interpret without the link to performance. As a vice president of IS remarked to us, "Is it good or bad if IS expenditures per employee are increasing?" Measures such as staff counts, or budget breakdowns, are useful only if clearly associated with measures of organizational impact. If we knew that increasing the development component of the IS budget was associated with increased revenue or decreased staff counts, then such data would be useful. On their own, they tend to be meaningless.

Linking IS measures to broader contexts of functional or corporate performance is an essential component of effective measurement. At present, while a considerable amount of information is collected by IS organizations, efforts linking these data to such broader contexts are largely missing. While poor measures and gaps in measurement undoubtedly limit management's ability to do so, it remains equally true that little effort is made to connect existing measures to larger contexts of understanding within IS or the organization. In short, as one senior IS manager noted, "We have lots of measures, but very little context."

Competitive Comparisons and Functional Benchmarking

Benchmarking, or comparing your organization to others, is a strategy that addresses many aspects of organizational performance. At a minimum, benchmarking can provide senior management with the assurance that the costs of their information systems function are not out of line with those of other organizations. But the true benefit of benchmarking is that it challenges managers to keep up with the best practices in business and to improve their productivity continuously. For example, finding out that Mazda could deliver accounting services using a fraction of the staff used at Ford, made Ford management realize that it had to rethink totally how they did this part of their business (Hammer, 1990).

Comparisons in IS are still rudimentary. Only one organization in the group undertook a formal benchmarking study on a regular basis (although others were investigating doing so). While almost all companies compare themselves to others, this is usually done in an informal manner. Some of the methods currently used include:

- **Asking around.** IS managers go to conferences and trade shows on a regular basis. Part of the benefit of these is to learn informally from the other participants about what they are doing and how successful it has been.

- **Visiting other Companies.** Many IS managers try to visit similar companies with whom they are not in competition to see what applications and practices they are using.

- **Vendors.** Vendors are often a good source of comparative information about the operation and deployment of their own equipment and software because of their knowledge of what goes on in other companies.

- **Joint Projects.** Some industries (e.g., travel) have the opportunity to collaborate on joint projects which enable participants to see firsthand how other companies work.

- **Industry Organizations.** Some industries have associations which collect and disseminate statistics on and to member organizations (e.g., the Life Office Management Association, LOMA).

- **Benchmarking Services.** Some companies hire professional benchmarking consultants to survey their organizations and compare them with others in the service's data base.

It is difficult to tell just how helpful these comparisons are. The problem with the first four of these methods is that they are so highly anecdotal that, as one IS manager stated, "No-one really knows who's winning." What are the factors associated with superior performance? It is difficult to compare oneself to industry leaders because there is no clear way of identifying who the leaders are and what made them that way.

Industry organizations and benchmarking services try to rectify this situation by collecting quantitative data for comparison purposes. While these services provide useful information, they each have certain limitations. Industry organizations are limited to single industry members inhibiting functional comparisons in other types of companies (see below). Benchmarking services are limited by the scope and quality of their data bases. For example, the service used by one group member had only three Canadian firms in their data base.

Comparisons and benchmarking let companies "know where they stand" in the larger business community. IS organizations could benefit substantially by undertaking more formal comparisons with other companies. While many problems with comparative data do exist (especially problems with imprecise definitions), there are few that could not be addressed if IS management were to manage measurement practices as carefully as they do other aspects of IS work.

Essential Principles of Effective Measurement

IS managers are more in agreement about what they would like to see. Group members came to a general consensus about the features of an effective measurement and benchmarking program. Together, these should form the basis of any organization's efforts to compare themselves internally or externally.

1. **Relevance.** One survey of CEOs and other top managers (IS and business alike) revealed that when assessing IS performance, they are seeking answers to three fundamental questions (Wilson, 1993):
   a) How well are we doing the things we are doing now?
   b) Are we doing the right things?
   c) Are we positioned to compete in the future?

   The most important test for any measure is whether or not it helps to answer one of these questions.

2. **Use of the Same Yardstick.** One of the benefits of benchmarking services, according to one manager, is that they force everyone to use the same set of measures. These then enable comparisons to be made. IS managers tend to see many of the problems associated with using common measures (e.g., problems with common

definitions, poor data collection) and none of the advantages. While such measures may be imperfect, the use of common measures not only makes comparison possible but opens the door to greater and greater refinements of these measures, thus improving the quality of measurement for all.

3. **Functional Benchmarking.** In the past, when companies undertook external comparison, they focused on their direct competitors. This was difficult since competitors are unlikely to share information with each other. Today, there is new recognition that value can be gained from comparing similar functions in a variety of different industries; and this information is much easier to obtain. For example, companies have compared order-taking processes from several industries (Wiesendanger, 1992). While there are many differences in the process of ordering a pizza and that of ordering cable TV services, there are also many similarities and much to be learned. IS itself is one function that can be benchmarked in this manner. While there are many differences in applications, the processes involved in developing, installing, and operating those applications are fundamentally the same regardless of the industry. It is thus an ideal candidate for functional benchmarking.

4. **Internal and External Comparisons.** As the Baldridge criteria note, it is important to assess the IS function in both ways. Certain measures, such as network availability, or benefits achieved, are essential for monitoring performance and quality internally. Others, such as descriptive statistics on costs and headcounts, are more appropriate for external comparisons. Finally, some measures, such as customer satisfaction or development effectiveness, can be used in both ways.

5. **Canadian Focus.** There are enough significant differences in uses of information technology and the business climate in Canada to warrant national comparisons of IS first and foremost. To our knowledge, Canadian firms have very few, if any, opportunities to compare themselves against other Canadian companies in any systematic way.

6. **Future Focus.** While statistics can assure management that it is operating efficiently and effectively in the past and present, the future dimension should never be forgotten. While it would be impossible to measure the future, interpretation of current data, as well as data-gathering, should not disregard analysis for the future. This is where many current practices of trend analysis fit in. Documenting and assessing industry-specific, IS practice, and technical trends, and how well your company is positioned vis à vis these, is an essential component of measurement.

However, managers should be warned against too-facile acceptance of "what everyone says is coming." One IS vice president noted that one survey showed substantial agreement amongst IS managers about how they would be managing in the year 2000. The only problem was that no one knew how to get there from where they were at present! While visions for the future are important, practical assessments about what to do in the short and medium term and how to evaluate success, are a necessary measure for companies and their management.

Six Types of Effective Measurement

Effective measurement for IS requires the use of appropriate tools designed to support each other. Good data are the result of properly-designed measures; that is, they are well defined and reflect what they are supposed to measure. In IS, different types of measures provide different types of information to management.

1. **Contextual Measures.** Since IS measurement is an art, not a science, understanding as much as possible about the organization and the management of a company and its IS function is essential to the proper interpretation of data. Contextual data provide important clues as to the reasons for certain findings. They assist the analyst in interpreting findings and determining their relevance to the organization. (Some examples of

contextual measures are given in Appendix A.)

2. **Diagnostic Measures.** Rather than collecting a complete cornucopia of data on a regular basis, diagnostic screening measures give a general picture of IS performance and quality in key areas. Just as certain medical tests screen for a range of health problems, in IS certain measures of quality and performance can highlight problem areas and the need for further study. The use of a series of diagnostic measures on a regular basis would eliminate the need for the massive on-going data collection activities that are often a significant part of IS work. (Some suggested diagnostic measures are listed in Appendix B.)

3. **Analytic Measures.** Where problems or superior performance are identified by diagnostic measures, more detailed analytic measures can be applied to discover the reasons why. Analytic measures focus on particular areas, and while detailed, are only used at certain times, rather that on a regular basis. They focus a spotlight on one area of the organization for a limited period of time only. When performance improves, analytic measures are disbanded. One organization has used this approach successfully in tuning its production systems to run more efficiently. They focus detailed analytic measures on one system at a time to achieve substantial operating improvements. When these are achieved, attention shifts to other systems. Analytic measures can be varied according to the problem. Because such data are short term, they can and should be organization and situation specific.

4. **Longitudinal Measures.** Any IS manager can tell a dozen "war stories" of the business problems caused by IS. What they also recognize, however, is that most of these problems are short-term only. Projects can get delayed at a point in their schedule and still finish on time. Systems can experience problems on installation, causing users to curse and gnash their teeth, and still benefit the company in the long term. Companies that invest heavily in one large system may show no return on their investment until the system is fully operational. Thus, measures of performance and quality that rely only on one "snapshot" of the organization, can be subject to numerous errors of interpretation. In order to capture a true picture of IS performance, it is necessary to have measures that represent a series of pictures over time. Only then will trends and consistent performance be apparent.

5. **Intuitive Measures.** Effective measures are those that others can intuitively understand. IS organizations have a tendency to create new and esoteric measures that senior management cannot relate to anything in their experience. Intuitive measures make this connection naturally for the reader. The best measures are therefore those that are expressed in terms well-known in business, such as "sales" or "revenues." More complex measures should be left for detailed analysis only.

6. **Qualitative and Quantitative Measures.** Statistics can only tell part of any story. Illustrative stories, quotations, explanations, descriptions, opinions, and feelings each have a place in assessing IS performance. While statistics are important measures because they lend force and credibility to any assessment, an understanding of what these figures mean can best be gained by qualitative support. Managers are also cautioned that many so-called quantitative studies do not capture objective reality but reflect subjective feelings, or worse, are designed to reflect back what the survey authors want to hear. Client surveys are a good example of this. Qualitative supporting material for statistics such as "67% of users stated they are satisfied with IS services" could help illuminate what IS is doing well and where it still needs to improve. Very often statistics leave out much of what can be learned. Ideally, therefore a mix of measures should be used, particularly where such "soft" measures as customer satisfaction or system performance are involved.

An Action Plan for Effective Measurement

While most IS organizations claim they want effective measurement, many simply are not prepared to put the effort and "think time" into doing it properly. Effective measures need not be time consuming to collect (indeed they should not), but they do require refocussing of attention about what to collect and careful thought about how to interpret the findings. These are the facets of measurement that are most often given shortest shrift in IS today. For managers short of time, professional researchers or measurement services can be employed, but participation in the process is essential. If your organization truly wishes to create effective measurement practices, the following action plan should be your guide:

1. **Make a Commitment.** As noted above, much of the best information is derived from longitudinal data, collected over a series of years so trends can be perceived. Management must make a commitment to collect data in the beginning even if it does not realize substantial benefits immediately. A commitment to accuracy must be made as well. The "garbage in, garbage out" maxim applies in this field, as in many others.

2. **Select and Develop Measures.** Many IS managers have deplored the lack of good measures that accurately reflect performance and quality. Yet IS managers are in an excellent position to develop better ones. Look for measures that relate directly to what you want to discover. Test them with the people who will be providing the data, and be prepared to make revisions. (You seldom get everything right the first time.) Assess the validity of measures by collecting qualitative data in addition to statistics. One organization began to poll all of its outlets on a random basis to see whether their systems are working, because traditional system availability figures contrasted with user complaints of system downtime. This measure has yielded useful information on system availability and in turn has developed into a much more meaningful measure than those generated by operations.

3. **Focus on Diagnostics.** Rather than measuring everything, managers should focus on creating and capturing diagnostic information. This will reduce the time involved in data collection. Management should look for and ensure that relevant measures are developed. (See Appendix B for a suggested list of these measures.)

4. **Look for Gaps.** Make sure your measurement program covers the five key areas identified by the Baldridge criteria, namely: customers, products and services, internal operations and performance, suppliers, and cost and financial information.

5. **Share with Other Companies.** Look for opportunities to work with a other companies to share common data and to develop industry standards for measurement. Business and industry organizations (e.g., The Conference Board, CIPS, ITAC, LOMA), benchmarking services, and universities are all doing work in this area.

6. **Integrate Findings.** Much information is lost due to the failure to assess carefully the data that are already collected in IS. Rather than a series of isolated reports on various subjects (e.g., customer satisfaction, comparison with other organizations, descriptive statistics), try to link all information together with the three questions senior managers are asking (see above). Then, try to describe what is known from a variety of sources under these broad general questions. For example, in addressing the question, "Are we doing the right things?", look at corporate performance measures as they relate to IS, what customer surveys say, and trends in key diagnostic measures over time. The more data can be integrated, the more you will find that the context of the results will appear. Anomalies and corroborative evidence will also become obvious.

7. **Interpret Results.** The interpretation of the findings in any measurement program is critical. One manager expressed concern that measurement results were often used to divide employees rather than motivate them. Any suggestion that a measurement

program will be used to evaluate individuals or even teams will ensure that people will find ways to subvert the process. Ideally therefore, findings should be kept at the level of the IS function (i.e., the whole function, as well as development and operations), rather than broken down by team or individual. Areas should be challenged to pull together and support each other, discouraging finger-pointing.

A second facet of interpretation concerns understanding negative findings. It is natural to react to such results with rationalizations and justifications about how one's own organization is "different." Such reactions will defeat the purpose of any measurement program. While it is appropriate to explore if errors in understanding or measurement were made, managers should also be asking questions about what they can learn from the findings.

8. **Analyze.** While measurement programs do reassure senior management that their IS organization is well-managed, the true benefits to the company as a whole come from the problem areas that are discovered. The analysis and resolution of problems is therefore essential to an effective measurement program. Analysis of most problems should have both a quantitative and a qualitative dimension, and should be addressed in a positive, non-judgemental manner. Where benchmarking has revealed significant divergence from best practice, site visits should be arranged. Again, keep in mind what can be learned from different industries. Detailed measurements should be undertaken both to explore the problem and to reassure management that it has been effectively addressed.

Conclusion

This paper has addressed the topic of IS measurement. IS managers are committed to measurement for many reasons: organizational credibility, effectiveness, efficiency, and general improvement. While some may question the benefits or accuracy of certain measures, most IS managers do not want to eliminate measurement from their organizations. Instead, they want better information — more accurate, easier to collect, better interpretations, and clearer definitions. In spite of all we don't know about how to measure IS, each of these is achievable at present. While there are gaps in measurement practices, these are fewer than most managers realize. The true key to effective measurement is quality, not quantity. Making use of the data already available (with judicious pruning and additions), more effective integration and analysis, as well as a long-term commitment to internal and external measurement, will yield results. Like so many other things in IS, good measurement derives from good management rather than a single magic answer.

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Appendix A: List of Measures Currently Used in IS

I. Corporate Context

- Company size (assets, customers, employees, locations)
- Industry
- Head Office Size
- Sales
- Sales/Employee
- Competitive Environment
- Organization Structure
- Reporting Structure
- Morale Index

II. Overall IS Performance

- IS Costs
- IS Costs/Corporate Revenues
- Technology Costs/Corporate Revenues
- Total Benefits Achieved
- Total Sales/ MIP
- Total Sales/ GB
- IS Salary Cost / Sales
- Time to fill hardware/software orders
- User Satisfaction (competitiveness, expert support, fulfilment, reliability, responsiveness, ease of doing business, partnership, overall satisfaction)
- Achievement of Service Agreements
- End User Support Survey
- Executive Satisfaction
- Employee Satisfaction

III. Descriptive Information About IS

A. IS Costs

- IS Budget / Net Sales
- IS Budget - Actual Costs
- IS Budget / Total Expenses
- IS Salaries - Permanent, Contact Staff
- Overtime Costs
- Cost of Benefits
- Training Costs
- Total Staff Cost / IS Budget
- Hardware Costs
- Hardware Maintenance Costs
- Software Purchase Costs
- Software Rental Costs
- Communications Costs
- Disaster Recovery Costs
- Total Equipment Cost
- Capital Expenditures
- Expenditures/IS Employee
- Expenditures/Company Employee

B. Human Resources

- Total Staff
- Total Contractors
- Total Management and Administration
- Total Development Staff
- Total Operations Staff
- Total Technical Services Staff
- Total Staff / Total Employees
- Skilled Staff (numbers trained to use CASE, methodology, GUI, Code Generators)
- Tool Usage (numbers using CASE, methodology, GUI, Code Generators)
- Staff Hours (charged, not charged, by phase, development, maintenance)
- Hours of Training
- New Hires
- Terminations
- Absences
- Moves
- Promotions
- Resignations
- IS Skills
- IS Job Descriptions

C. IS Context

- Reporting Structure
- Application Packages Used (number, variety)
- Hardware Platforms (number, variety)
- Tools Used
- Portfolio Assessment (age, complexity, platform)
- Degree of Centralization (processing, report distribution, location of help, staff, LANS)

IV. Operations Performance

- Number of MIPS
- GB DASD
- Lines Printed
- CPU Hours
- Equipment Installs, Moves, Removes
- Capacity Utilization (CPU, DASD, tape, printers)
- Workload /MIP (batch, online, interactive, DASD, output, time of day)
- Response Time (network, CICS, TSO, VM)
- System Availability Poll
- Availability Levels By Equipment, Network
- Production Hours / Year
- Turnaround Time (batch)
- Meantime to Recover (minutes/outage)
- Cost Allocations (CICS, TSO, batch, DASD, printing, forms, leases, people, testing, telecommunications, equipment)
- CPU Cycles / Business Transaction
- Help Desk Productivity (calls handled, total calls, length of call, by time of day, abandoned calls, time to answer)

V. Development Performance

- Function Points / Work Month
- Delivery On-Time Trends
- Deliverables (completed, not complete, not started) for service requests, new projects, production problems
- Function Points / Person (cost, for in-house development, for packages)
- Time to Market
- Development Budget / Maintenance Budget

VI. Project Information

A. Project Performance

- Benefits Delivered
- Function Points Delivered
- ROI
- Impact on Market Share
- Impact on Strategic Direction
- Work Value Assessment
- Alignment with Business Objectives
- Number of Changes
- System Performance
- Number of Abends and Reason
- Production Hours / Year

- Number of Work Requests
- Number of Hours on Post-Release Problems
- Number of Changes / 100 Function Points
- Defects / 100 Function Points
- Staff Months / Function Point

B. Project Descriptive

- Project costs
- Actual Costs / Budget
- Progress against schedule
- Training costs
- Paid hours / work unit
- Manmonths / Project

Appendix B: Some Suggested Diagnostic Measures

Are We Doing Things Right?

- Total Benefits Delivered (user estimate) — IS overall, project [Cost & Financial]
- Cost/Transaction [Internal Performance, Cost & Financial]
- Average Time to Market [Internal Performance]
- Project Progress against Schedule/ Budget [Internal Performance]
- Delivery Time for Hardware/ Software [Supplier Performance, Internal Performance]
- Number of Calls to Help Desks [Product Performance]
- System Availability Index [Service Performance]
- Number of Abends, Number of Work Requests [Product, Service Performance]
- Customer Satisfaction Index [Customer]

Are We Doing the Right Things?

- Business Revenues /Employee (over time) [Cost & Financial]
- Return on IT Investment vs. Return on HR Investment [Cost & Financial]
- Executive Satisfaction Index [Customer]
- Trends in Cost/Transaction, Time to Market, Hardware/Software Delivery Time, Calls to Help Desk, System Availability Index, Number of Abends, Number of Work Requests, Customer Satisfaction Index over Time.

Are We Positioned for the Future?

- Competitive Environment [Customer, Product & Service Performance, Supplier Performance]
- Commitment to Research and Development [Customer, Product & Service Performance]
- Portfolio Assessment (function, age, complexity, platforms) [Product & Service, Customer]
- IS Costs, IS Returns, Cost/Transaction compared to Best Practices [Cost & Financial, Internal Performance]